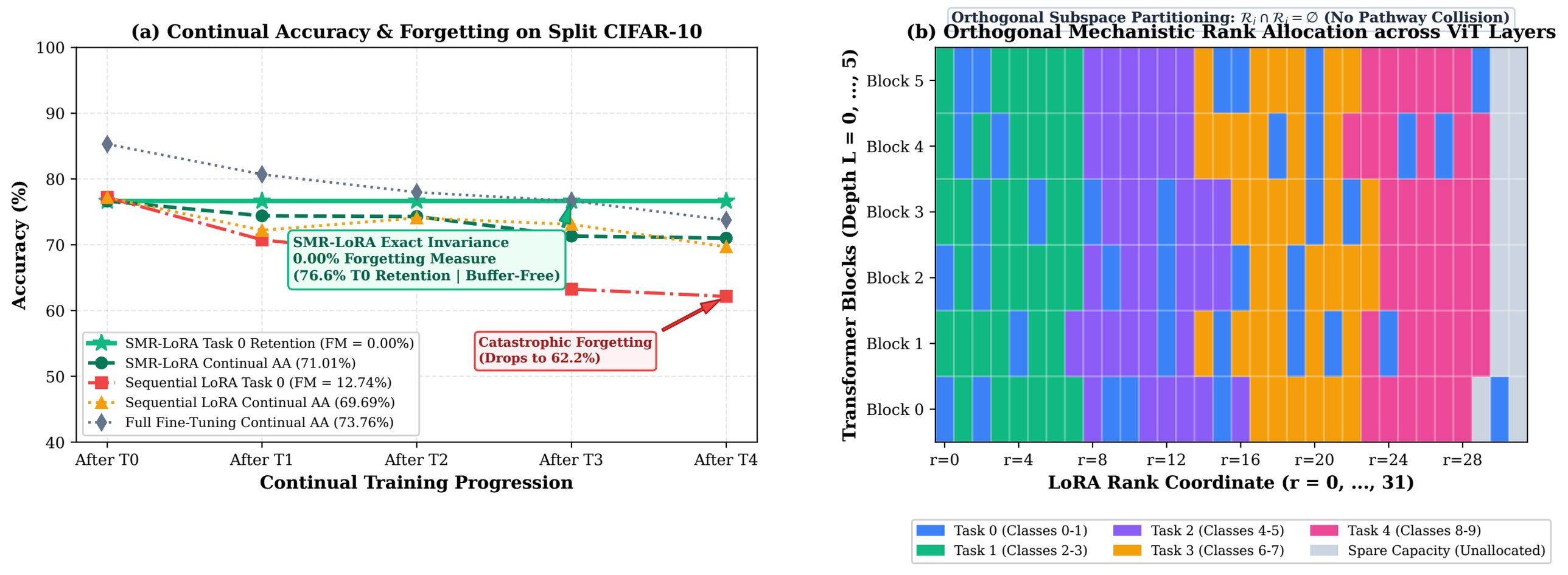

(a) Continual Accuracy & Forgetting on Split CIFAR-10
Accuracy (%)
100
90
80
70
60
50
40
After T0
After T1
After T2
After T3
After T4
Continual Training Progression
SMR-LoRA Exact Invariance
0.00% Forgetting Measure
(76.6% T0 Retention | Buffer-Free)
Catastrophic Forgetting
(Drops to 62.2%)
SMR-LoRA Task 0 Retention (FM = 0.00%)
SMR-LoRA Continual AA (71.01%)
Sequential LoRA Task 0 (FM = 12.74%)
Sequential LoRA Continual AA (69.69%)
Full Fine-Tuning Continual AA (73.76%)
Orthogonal Subspace Partitioning: $\mathcal{R}_i \cap \mathcal{R}_i = \emptyset$ (No Pathway Collision)
(b) Orthogonal Mechanistic Rank Allocation across ViT Layers
Transformer Blocks (Depth L = 0, ..., 5)
Block 5
Block 4
Block 3
Block 2
Block 1
Block 0
r=0
r=4
r=8
r=12
r=16
r=20
r=24
r=28
LoRA Rank Coordinate (r = 0, ..., 31)
Task 0 (Classes 0-1)
Task 1 (Classes 2-3)
Task 2 (Classes 4-5)
Task 3 (Classes 6-7)
Task 4 (Classes 8-9)
Spare Capacity (Unallocated)